Consumption of whole grain reduces risk of deteriorating glucose tolerance, including progression to prediabetes.

BACKGROUND: High whole-grain intake has been reportedly associated with reduced risk of developing type 2 diabetes (T2D), which is an effect possibly subject to genetic effect modification. Confirmation in prospective studies and investigations on the impact on prediabetes is needed. OBJECTIVES: In a prospective population-based study, we investigated whether a higher intake of whole grain protects against the development of prediabetes and T2D and tested for modulation by polymorphisms of the TCF7L2 gene. DESIGN: We examined the 8-10-y incidence of prediabetes (impaired glucose tolerance, impaired fasting glucose, or the combination of both) and T2D in relation to the intake of whole grain. Baseline data were available for 3180 women and 2297 men aged 35-56 y. RESULTS: A higher intake of whole grain (>59.1 compared with <30.6 g/d) was associated with a 34% lower risk to deteriorate in glucose tolerance (to prediabetes or T2D; women and men combined). The association remained after adjustments for age, family history of diabetes, BMI, physical activity, smoking, education, and blood pressure (OR: 0.78; 95% CI: 0.63, 0.96). Risk reduction was significant in men (OR: 0.65; 95% CI: 0.49, 0.85) but not in women. Associations were significant for prediabetes per se (all, OR: 0.73; 95% CI: 0.56, 0.94; men, OR: 0.57; 95% CI: 0.40, 0.80). The intake of whole grain correlated inversely with insulin resistance (HOMA-IR). The impact of whole-grain intake was undetectable in men who harbored diabetogenic polymorphisms of the TCF7L2 gene. CONCLUSIONS: A higher intake of whole grain is associated with decreased risk of deteriorating glucose tolerance including progression from normal glucose tolerance to prediabetes by mechanisms likely tied to effects on insulin sensitivity. Effect modifications by TCF7L2 genetic polymorphisms are supported.